

Writing Large Numbers in Scientific NotationQuestion Bank - Additional 3.08 **ANSWER KEY**

Show your work. Circle your final answer.

For questions 1 through 4, write each number in scientific notation.

1. 70,000

$$7 \times 10^4$$

2. 2,800,000

$$2.8 \times 10^6$$

3. 640,000,000

$$6.4 \times 10^8$$

4. 5,030,000,000

$$5.03 \times 10^9$$

For questions 5 and 6, write each number in standard form.

5. 4.1×10^5

$$410,000$$

6. 1.64×10^7

$$16,400,000$$

Real quantities.

7. Write each quantity in scientific notation.

- a. Earth's beaches hold about 7,500,000,000,000,000 grains of sand.

$$7.5 \times 10^{18} \text{ grains}$$

- b. A distance of about 39,900,000,000,000 km.

$$3.99 \times 10^{13} \text{ km}$$

- c. A computer displays a population as **1.2E7**. Write it in standard form.

$$12,000,000$$

8. Complete each comparison.

- a. How many times larger is 8.2 than 4.1?

2

- b. How many times larger is 10^7 than 10^5 ?

100

- c. How many times larger is 8.2×10^7 than 4.1×10^5 ?

$2 \times 100 = 200$ times larger

9. Answer each question.

- a. About 6.94×10^6 people visited one theme park in a year, and about 1.7×10^6 visited another. About how many times larger is the first number? Round to the nearest whole number.

about 4 times

$6.94 \div 1.7 \approx 4.08$; the powers are equal.

- b. China's population is about 1.44×10^9 and Mexico's is about 1.292×10^8 . About how many times larger is China's? Round to the nearest tenth.

about 11.1 times

$1.44 \div 1.292 \approx 1.115$ and $10^9 \div 10^8 = 10$.

10. Which of the following is greater? Explain how you know.

- a. 8.7×10^4 or 2.1×10^5

2.1×10^5 — 10^5 is greater than 10^4 ; $210,000 > 87,000$.

- b. 5.6×10^6 or 5.09×10^6

5.6×10^6 — same power; $5.6 > 5.09$.

11. Which is 71,000,000 written in scientific notation?

☐ A. 7.1×10^8

☐ B. 71×10^6

☐ C. 7.1×10^6

☒ D. 7.1×10^7

D

12. Select ALL of the expressions that are equal in value to 6,200,000.

- ☐ A. 6.2×10^5
- ☒ B. 620×10^4
- ☒ C. 0.62×10^7
- ☐ D. 6.2×10^7
- ☒ E. 62×10^5
- ☒ F. 6.2×10^6

B, C, E and F

Only F is in scientific notation.

13. In question 12, only one of the four equal expressions is written in scientific notation. Which one is it, and why are the other three not?

6.2×10^6 . The coefficient must be at least 1 and less than 10; 62, 620 and 0.62 are not.